Effect of Planting Season, Bunchgrass Species, and Neighbor Control on the Success of Transplants for Grassland Restoration

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Abstract

Constraints to grassland and open forest restoration (e.g., poor seed sources, yearly variation in establishment, and the persistence of weeds) necessitate the development of innovative methods to restore bunchgrass communities. We assessed the use of two native bunchgrass transplants, Bluebunch wheatgrass (*Pseudoroegneria spicata*) and Spreading needlegrass (*Achnatherum richardsonii*), for restoration within thinned montane forest communities of southeastern British Columbia, Canada. Fall and spring plantings were examined, either with or without glyphosate treatments to Pinegrass (*Calamagrostis rubescens*) neighbors. *Calamagrostis rubescens* is abundant in grassland affected by tree encroachment and may limit transplant establishment. Bunchgrass survival was positively associated ($p < 0.05$) with transplant size. Although *P. spicata* survival was greater ($p < 0.01$) with fall (81%) than with spring (44%) planting, survival of *A. richardsonii* was greater ($p < 0.01$) when planted in the spring (68 vs. 23%). Reduction of *C. rubescens* led to a relatively small but significant increase ($p < 0.05$) in bunchgrass survival by 7%. The summer after planting, changes in transplant tiller number varied by bunchgrass species, planting season, and treatment of neighboring *C. rubescens*. Removal of neighboring *C. rubescens* generally increased the number of tillers (or reduced tiller loss) but only within fall-planted *A. richardsonii* and spring-planted *P. spicata*. Both *A. richardsonii* and *P. spicata* transplants have potential for understory restoration within thinned montane forests, particularly using larger individuals, although to maximize survival, these species should be planted in the spring and fall, respectively. Reduction of *C. rubescens* may also enhance transplant survival and in some cases growth.

Key words: **Bluebunch wheatgrass, bunchgrass mortality, conifer ingrowth, grassland restoration, Pinegrass, planting season, Spreading needlegrass, tillers.**

Introduction

Many grasslands have been encroached by woody species (Archer et al. 1988; Ueckert et al. 2001). Montane ecosystems of North America are no exception (Rhemtulla et al. 2002), caused in part by fire suppression in areas well adapted to this disturbance (Tande 1979; Johnson & Fryer 1987). Within the montane forests, this process is evidenced by widespread changes from fire-resistant, relatively open conifer stands containing an understory of bunchgrasses, to ingrown, closed canopy stands of shade-tolerant and fire-sensitive species (Cooper 1960; Lunan & Habeck 1973; Arno & Gruell 1986; Habeck 1990).

Increased competition for light from shade-tolerant species limits the abundance of native bunchgrasses that were common before tree invasion (Lunan & Habeck 1973; Bojorquez et al. 1990; Page et al. 2005). The retention and restoration of montane grasslands is important from the perspective of conserving biodiversity (Watkinson & Ormerod 2001), as well as in maintaining ecosystem function, including limiting potential carbon loss attributed to woody encroachment (Jackson et al. 2002) and providing forage for ungulates (Wikeem & Pitt 1992). Even with overstory thinning using prescribed fire or mechanical removal, the successful restoration of bunchgrass communities in the understory of ingrown forests may require active reestablishment of bunchgrasses using seeding or transplants.

Grassland restorations are often impeded by constraints such as a lack of reliable seed sources, yearly variation in weather, and the presence of competitive weeds, all of which affect establishment (Wilson 2002). Mulligan et al. (2002) reviewed studies documenting Wiregrass (*Aristida beyrichiana* Trin. & Rupr.) establishment for the purpose of restoring fire-maintained Longleaf pine (*Pinus palustris* Mill.) understory communities, and found germination rates to be unpredictable, frequently falling below 10%. Low germination is common in restoration seeding projects in arid and semiarid rangeland (Grantz et al. 1998).

Due to the limited success of seeding, transplants have been increasingly used for arid or semiarid environments including grasslands (Bainbridge et al. 1995; Grantz et al.

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1998; Davies et al. 1999; Ewing 2002; Mulligan et al. 2002) as well as for the restoration of riparian wetlands (Steed & DeWald 2003) and woodland (Sinclair & Catling 2003). Bypassing the seeding, germination and emergence phases of establishment are thought to increase restoration success. Transplants allow for rapid establishment of species that may take a long time to invade or establish through natural processes (Davies et al. 1999).

The restoration of ingrown montane forest to open forest habitats may require prescribed burning and silvicultural thinning (Page et al. 2005), after which the transplants could be used as a tool to initiate bunchgrass recovery. Preliminary tests indicate that planted nursery-grown grass plugs had 94% survival during the first year of monitoring (BC Ministry of Forests, unpublished data). Moreover, transplants may have better resistance to competition from established grasses such as Pinegrass (*Calamagrostis rubescens* Buckl.). *Calamagrostis rubescens* is a dominant rhizomatous grass of northern inland forests (Franklin & Dyrness 1973) and is abundant under dense conifer canopies (Steele & Geier-Hayes 1993). Pinegrass also spreads in response to light to moderate soil disturbance associated with mechanical site preparation or prescribed fire (Haeussler et al. 1990). Therefore, successful restoration of bunchgrasses may require active control of competition from *C. rubescens*.

Our objective was to experimentally test the survival and growth of bunchgrass transplants installed within the recently thinned forests undergoing active restoration. Additionally, transplant success was assessed with and without the control of neighboring *C. rubescens*. Although previous studies have evaluated the use of a preplanting broadcast herbicide treatment for enhancing grassland restoration (Stromberg & Kephart 1996; Ewing 2002), or selectively removed neighbors to assess interspecific interactions (Silander & Antonovics 1982; Peltzer & Köchy 2001), studies using the selective application of herbicide to reduce a specific competitive species in a grassland restoration context are less common.

Methods

Study Area

Planting trials were conducted at three blocks located within 100 km of each other in the Rocky Mountain Trench of southeastern British Columbia (lat 49°58′N, long 115°43′E; elevation 900–1000 m asl). This region has an upland continental climate with well-defined seasons (Marsh 1986). Mean monthly air temperatures vary from −8.3 to 18.2°C. Average annual precipitation is 385 mm, peaking in May and June. The summer prior to planting (2000) was unusually dry (45% of normal precipitation, May–September) as was the summer after planting (2001) (35% of normal precipitation, May–September).

Soils at the study sites were Orthic Eutric Brunisols, with a gravelly sandy loam as the dominant texture (Lacelle 1990). These soils are characterized by one or more organic surface horizons overlying a brownish, base-saturated Bm horizon. The C horizon is calcareous (National Research Council 1998).

All the blocks were similar in vegetation, being dominated by Interior Douglas-fir (*Pseudotsuga menziesii* (Mirb.) Franco ssp. *menziesii*), with *Calamagrostis rubescens* in the understory. Forests at each site were thinned in May or June 2000 from a mean of 504 to 243 stems/ha, to restore an open structure representative of conditions prior to ingrowth. Historically, these stands were maintained at low tree density through frequent wildfire and were dominated by bunchgrasses such as Bluebunch wheatgrass (*Pseudoroegneria spicata* (Pursh) A.Löve) and Spreading needlegrass (*Achnatherum richardsonii* (Link) Barkw.) (Hope et al. 1991). Current land uses of the area include cattle grazing. Sites were not protected from grazing during the planting trials.

Experimental Design

At each experimental block, bunchgrass transplants were planted 60 cm apart in rows during either the fall (8–9 October 2000) or the spring (8–11 May 2001), using a fully randomized design. Half the plants of each species at each planting date had neighboring *C. rubescens* controlled. Thus, the resulting treatments in each season of planting were (1) *A. richardsonii* with *C. rubescens* reduced; (2) *A. richardsonii* with *C. rubescens* untreated; (3) *P. spicata* with *C. rubescens* reduced; and (4) *P. spicata* with *C. rubescens* untreated. There was no supplementary watering or fertilizer application to ensure natural field conditions.

Neighboring *C. rubescens* was controlled using glyphosate (Monsanto Co., Muscatine, IA, U.S.A.), a translocated, nonresidual herbicide. Concentrated glyphosate solution (7 g/L) was applied by wiping the herbicide onto all *C. rubescens* shoots within 30 cm of the bunchgrass transplants using a small sponge. Transplants had a protective barrier placed over them at application to prevent exposure to herbicide. Translocation of herbicide to adjacent *C. rubescens* plants in untreated plots was not observed.

A total of 304 bunchgrass plugs were grown from seeds in a greenhouse at Tappen, British Columbia. All the seeds were collected within 100 km of the study sites. Three to four seeds were sown during July 2000 into containers of potting soil (80-mL volume, 4-cm diameter) and subsequently thinned to single plants. During October 2000, 40 plugs of each species were randomly selected and planted at each of the three blocks, with half receiving the herbicide treatment to neighboring *C. rubescens*. At fall planting, the transplants of *P. spicata* averaged 1.5 cm^2 ± 0.9 SD in basal area, 27 cm ± 5.7 SD in height, and had 9 tillers/plug ± 3 SD, whereas transplants of *A. richardsonii* averaged 1.3 cm^2 ± 1.2 SD in basal area, 8.1 cm ± 3.4 SD in height, and had a tiller count of 13 tillers/plug ± 6 SD. During the following spring (May 2001), 16 additional plugs of each species were planted at each of the two blocks, with half receiving the *C. rubescens* removal treatment. At the time of spring planting, transplants of *P. spicata* averaged 2.1 cm^2 ± 2.0 SD in basal area, 22.5 cm ± 7.5 SD in height, and had 4 tillers/plug ± 3 SD, whereas transplants of *A. richardsonii* averaged 4.9 cm^2 ± 2.4 SD in basal area, 20.3 cm ± 4.9 SD in height, and had 13 tillers/plug ± 5 SD. Sample sizes were reduced in spring due to a finite supply of plugs. Plugs held overwinter were kept cool (4°C) and at constant day length. Transplants were not root bound at any time.

In both seasons, transplants were randomly assigned to the *C. rubescens* removal treatment. Neighboring *C. rubescens* shoots around the fall transplants were treated a second time during spring planting to ensure optimal control. Plant height, tiller numbers, and basal area were assessed within each transplant at the time of planting. Both spring and fall transplants were measured for survival, evidence of grazing, and the number of tillers and inflorescences in September 2001.

Statistical Analyses

To meet the assumptions of analysis of variance (ANOVA), all data were tested for normality using Proc Univariate in SAS (SAS 1999), with no transformations necessary. The effects of planting season, bunchgrass species, and *C. rubescens* control on plug survival were tested using a split-plot design, with season of planting being the main plot factor and individual blocks as replicates (Table 1). To assess survival, the proportion (%) surviving was first calculated for each group of transplants within each unique combination of planting season, bunchgrass species, and *C. rubescens* reduction treatment, resulting in 20 observations with no subsampling. Initial mean tiller number per treatment group was incorporated as a covariate in the analysis of survival. To evaluate the direct effects of initial plant size on survival, a post hoc one-way ANOVA was used to compare the morphological characteristics of plants that survived and died. For the latter analysis, each bunchgrass species was examined separately due to their different morphological characteristics.

To assess tiller growth, an initial ANOVA using a split-plot design, similar to the survival analysis but with transplants as subsamples, was conducted on surviving plugs to determine the effect of planting season, bunchgrass species, and *C. rubescens* reduction on relative changes in the number of tillers between the planting and the end of the 2001 growing season (Table 1). Tiller data were assessed as the proportional change in tiller numbers to adjust data for nonuniform plant sizes. For this analysis, block effects were included but are only discussed where they provide unique information.

All results are based on the use of least squares means, with a Tukey–Kramer adjustment performed when conducting mean comparisons of significant effects. All results are considered significant when $p < 0.05$, although the main effect of *C. rubescens* removal was considered a one-tailed test due to the expectation that neighbor-vegetation control would enhance transplant survival and growth.

Results

Herbicide treatment significantly reduced ($p < 0.001$) the cover of *Calamagrostis rubescens*. In September 2001, plots treated with glyphosate had 47% less *C. rubescens* cover relative to untreated plots. *Calamagrostis rubescens* control was slightly more effective in the spring because spring-controlled plots had an average of 11% ± 15 SD cover, whereas fall-controlled plots had an average of 16% ± 15 SD. There was also variation in *C. rubescens* cover among the blocks, which ranged from 6.4 to 23.5%. Limited grazing occurred on the transplanted plugs, with the proportion of transplants grazed during 2001 varying from 2 to 15% among blocks.

Table 1. ANOVA summary table of the main effects of planting season, bunchgrass species, and *Calamagrostis rubescens* neighbor control on the survival and growth (change in tiller numbers) of transplants the growing season after planting.

	Survival (% of Plugs)				*Growth (Proportional Δ in Tillers/Plug)*			
Source	*df*	$\overline{X}$ *Square*	*F*	*Pr > F*	*df*	$\overline{X}$ *Square*	*F*	*Pr > F*
Initial covariate (tiller no.)				0.91				
Season	1	64.53	0.09	0.78	1	0.22	0.01	0.92
Block (season), error 1	3	693.06	11.17	<0.01	3	15.25	18.72	<0.001
Bunchgrass	1	1484.03	23.91	<0.001	1	0.00	0.00	0.99
C. rubescens	1	208.03	3.35	<0.05	1	1.60	1.97	0.08
Bunchgrass × *C. rubescens*	1	8.53	0.14	0.72	1	0.37	0.45	0.50
Season × bunchgrass	1	8300.03	133.75	<0.0001	1	10.40	12.77	<0.001
Season × *C. rubescens*	1	28.03	0.45	0.52	1	0.27	0.33	0.57
Season × bunchgrass × *C. rubescens*	1	22.53	0.36	0.56	1	3.44	4.23	0.04
Error 2	9	69.06			139	0.87		

p values for the main effect of *C. rubescens* control were adjusted for a one-tailed test. For change in tiller numbers/plug, the interactions of block with the main effects (data not shown) were significant only for block × bunchgrass ($p = 0.01$).

Bunchgrass Survival

Bunchgrass transplant survival varied among species ($p < 0.001$; Table 1), with a greater proportion of *Pseudoroegneria spicata* transplants (74%) surviving compared to *Achnatherum richardsonii* (30%). A significant season × bunchgrass interaction also occurred ($p < 0.0001$) due to the differential survival of species among planting seasons (Fig. 1A). Survival of *P. spicata* was greater ($p < 0.01$) when planted in the fall (81%) compared to that in the spring (44%). The opposite pattern was evident with *A. richardsonii*, with the survival of spring transplants (68%) greater ($p < 0.01$) than that of the fall transplants (44%).

Controlling *C. rubescens* significantly increased the overall survival of bunchgrasses ($p < 0.05$; Table 1). Although these effects ($p < 0.05$) were consistent for both the species (Fig. 1B), the extent of the change was limited, with survival increasing by 7% across both *A. richardsonii* and *P. spicata*.

The initial size of transplants, as represented by tiller number, basal area, and height, varied ($p < 0.05$) between the plugs that survived and the plugs that died during the study (Table 2). With the exception of tiller numbers for *A. richardsonii* and height for *P. spicata*, this trend was consistent for both bunchgrass species. Larger transplants had a greater likelihood of survival during the first growing season after establishment (Table 2).

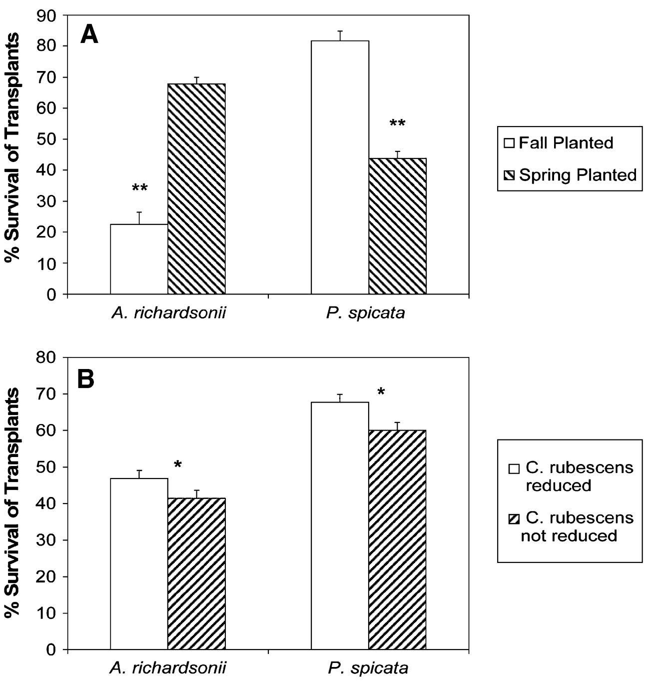

Figure 1. Effect of season of planting (A) and *Calamagrostis rubescens* neighbor control (B) on least squares means (±1 SE) survival (%) of two species of bunchgrass transplants in 2001. Within a species, * and ** indicate a significant difference at $p < 0.05$ and $p < 0.01$, respectively.

Bunchgrass Growth

Among the surviving transplants, observed changes in bunchgrass transplant growth as reflected by relative changes in tiller numbers, were affected by the treatments, which included a significant ($p < 0.001$) season × bunchgrass interaction and a significant ($p = 0.04$) season × bunchgrass × *C. rubescens* interaction (Table 1). Results of the latter three-way interaction are provided in Figure 2. *Achnatherum richardsonii* transplants that survived fall planting lost fewer tillers ($p = 0.04$) compared to those surviving spring planting. In contrast, *P. spicata* transplants that survived spring planting lost fewer tillers ($p = 0.002$) than those surviving from the previous fall.

Calamagrostis rubescens control had contrasting effects on transplant tiller growth. Reduction of *C. rubescens* resulted in a positive response in tiller numbers but only within the *A. richardsonii* transplants surviving from the fall and *P. spicata* transplants surviving from the spring (Fig. 2). Tiller growth in all other treatments was typically negligible or negative, with no improvement resulting from neighbor control.

Only a single block-based interaction was found in the analysis of tiller data, that being a significant ($p = 0.02$) block × bunchgrass effect. Between species, *P. spicata* plugs lost a greater number of tillers (−34%) versus *A. richardsonii* plugs (+6%). *Achnatherum richardsonii* outperformed *P. spicata* (i.e., either gained more or lost fewer tillers) at all three study sites, and the difference between them ranged from 14 and 16% to a high of 66%. Thus, the advantage of *A. richardsonii* was particularly apparent at one of the three study sites.

Although inflorescence data were too variable to detect significant differences among treatments, it is noteworthy that all the plugs producing seed heads during the study were planted in the fall (32 of 240). *Pseudoroegneria spicata* comprised 30 of these 32 plugs. Seed heads were not examined for the presence of viable seed.

Discussion

Suitability of Transplants for Bunchgrass Establishment

Transplant survival varied with all factors studied, including bunchgrass species, planting season, and competition from *Calamagrostis rubescens*. Bunchgrass survival was also affected by the initial size of the transplant. Although more than 50% of transplants survived the first growing season, survival ranged widely (23–83%) among treatments, suggesting that the success of bunchgrass transplants in restoration is variable. Moreover, given the reduction in tiller number after planting, the ability of transplants to remain in these communities over the long term is questionable.

Nevertheless, because success in restoration is defined by the outcome of specific actions relative to project objectives, even a low level of bunchgrass establishment may be considered successful. In this study, surviving transplants reestablished a low density of both *Achnatherum*

Table 2. Comparison of mean initial morphological characteristics (±1 SD) at the time of planting between bunchgrass transplants that subsequently survived and bunchgrass transplants that subsequently died during the 2001 growing season.

Characteristic	*Species*	*Live Transplants*	*Dead Transplants*	*Pr > F**
Basal area (cm^2)	*Achnatherum richardsonii*	2.8 ± 0.8	1.5 ± 0.3	0.02
	Pseudoroegneria spicata	1.7 ± 0.5	1.0 ± 0.7	0.002
Height (cm)	*A. richardsonii*	14.2 ± 7.4	9.5 ± 4.6	<0.001
	P. spicata	27.0 ± 5.8	25.2 ± 7.8	0.07
Tiller number (no.)	*A. richardsonii*	12.3 ± 4.0	13.1 ± 5.2	0.38
	P. spicata	8.1 ± 3.5	6.4 ± 3.9	0.009

**p* values compare means between live and dead plugs within a row.

richardsonii and *Pseudoroegneria spicata* and could therefore serve to accelerate recovery of these bunchgrass communities. However, the decision on whether the cost and effort associated with the level of establishment achieved is justified will ultimately depend on the perceived importance of bunchgrass restoration to land managers and their land use objectives.

Interpretation of our results should be tempered by consideration of the growing conditions during 2000 and 2001, the years immediately preceding and after planting, respectively. Dry weather combined with other stresses may have contributed to the low and variable seasonal and interspecific variations in bunchgrass survival and tiller growth. Although grazing has been implicated elsewhere in reducing grass transplant survival (Best and Bork 2003), this was not considered a factor here due to low grazing pressure ($\leq$15% of plants were defoliated). Although our results suggest that transplants may be used for restoration of thinned dry montane forests, even during the extended droughts common within the region, planting during moister periods could enhance survival. Alternatively, moister conditions may favor greater growth of competing vegetation including *C. rubescens*, in turn reducing bunchgrass survival and growth.

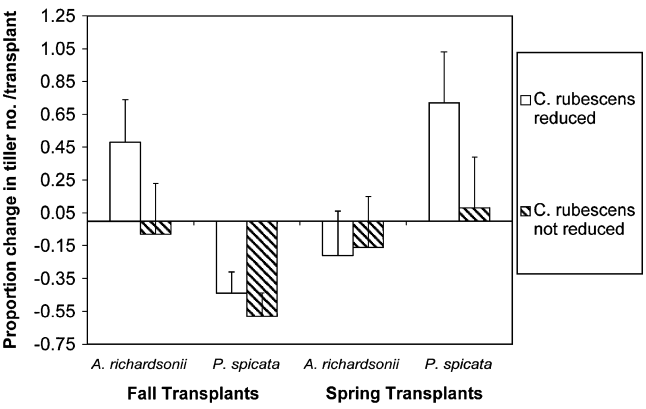

Figure 2. Proportional change in plug least squares means tiller number (±1 SE) during the 2001 growing season for various combinations of planting season, bunchgrass species, and *Calamagrostis rubescens* neighbor control.

Species Differences

Our results indicate that establishment success is related to species differences. Survival was optimized by planting *P. spicata* in the fall and *A. richardsonii* in the spring. *Pseudoroegneria spicata* is a cool-season (C_3) species that initiates growth early, often by February in this region (Willms et al. 1980). We hypothesize that although fall planting may allow effective acclimation of *P. spicata* to the new environment, planting plugs of this species in spring (i.e., May) could shorten its growing season and promptly expose transplants to moisture stress, decreasing their survival. In contrast, although *Achnatherum* is also a cool-season species, like other needlegrasses, *A. richardsonii* appears to be affected more by temperature (Parsons et al. 1971; Gurevitch 1986) than by length of the growing season. Thus, this species may be better able to establish when planted during the warmer conditions of spring, shortly before active growth.

Although planting season had a clear effect on survival, the effects of season on tiller growth were not as obvious. Moreover, trends in transplant growth based on tiller numbers were opposite those of survival. *Achnatherum richardsonii* plugs planted in the fall produced more or lost fewer tillers than *P. spicata* transplants, with the opposite true with spring planting. This dichotomy may be partly due to indirect selection for strong growth traits through plug mortality because transplants that survived in suboptimal planting conditions may have had traits that predisposed them to superior growth, resulting in a bias among surviving plugs toward greater tiller increases (or fewer tiller losses). These results were further supported by the established selection bias within plugs that survived planting in favor of larger individuals. Other studies using transplants have found that initial plant size may confer a competitive advantage (Davies et al. 1999; Steed & DeWald 2003), although this notion has not held true elsewhere (Fehmi et al. 2004). Davies et al. (1999) found that transplant size was only significant during the initial period after planting and the early establishment phase. Our findings suggest that where feasible, larger transplants be used to enhance bunchgrass establishment during the initial phase of grassland restoration, although this benefit will have to be evaluated relative to any added cost.

Although inflorescence production was limited in this study, *P. spicata* transplants, particularly those fall planted, produced some seed heads by the end of the first growing season. Although seed head presence is not necessarily indicative of viable seed availability, even low seed production is important in areas undergoing restoration by increasing the likelihood of future seedling recruitment. This response is particularly important for bunchgrasses because it represents a key recovery mechanism (i.e., replenishment of the soil seed bank) within communities undergoing restoration. Notably, this occurred despite the decline in tiller numbers reported earlier and the occurrence of drought, suggesting that these perennial bunchgrass plants were able to cope with their new environment.

Benefits of Neighbor Control

The survival and vigor of transplants was increased by control of neighboring *C. rubescens*. Transplants with neighbor control experienced 7% greater survival regardless of bunchgrass species and the season of planting, which is similar to observations in other studies on the importance of interspecific competition (Brown & Bugg 2001; Fehmi et al. 2004). Competition for limited resources may determine the presence, absence, or abundance of species in a community and determine their spatial arrangement (Pyke & Archer 1991). Although transplant success may be determined in part by the size of the gap created through the selective removal of *C. rubescens*, only one gap size was examined here (30 cm). Other studies using similar bunchgrass species (Lemmon's needlegrass [*Achnaetherum lemmonii*] and *P. spicata*) found an increasing negative effect of smaller distances between competing vegetation, although influences were negligible above 18 cm (Huddleston & Young 2004). Thus, changes in the 30-cm gap size or alterations in the number of neighboring plant species present could have altered bunchgrass responses.

In the context of restoration, competition for resources is often discussed based on non-native and native species (Wilson 2002). In this case, *C. rubescens* is a native species but nonetheless a vigorous competitor in this environment, as exemplified by its tendency to increase with forest ingrowth at the expense of other species including bunchgrasses (Page et al. 2005). Based on our results, competition from *C. rubescens* had an adverse impact on plug survival regardless of planting season. *Calamagrostis rubescens* is a rhizomatous species that initiates growth early in the spring (McLean 1979) and has 53% of its biomass belowground in creeping rhizomes (Stout et al. 1983). With a shallow rooting habit and early emergence, *C. rubescens* is a highly effective competitor for the limited moisture found in dry *Pseudotsuga menziesii* stands during spring. Moreover, drought may have exacerbated the intensity of competition on transplants where *C. rubescens* was not reduced.

Competition from *C. rubescens* also limited transplant growth the year after planting, particularly fall *A. richardsonii* and spring *P. spicata* transplants. These results are similar to those from northwestern Montana where *C. rubescens* competition had a negative impact on *Pinus ponderosa* seedling stemwood, foliage, and root weight (Peterson 1988). Other studies have shown that early-emerging species, including exotic (i.e., non-native) grasses as well as *C. rubescens*, continually increase their ability to capture resources at the expense of later emerging species and, in doing so, increase their physical zone of influence (Ross & Harper 1972). As a result, when bunchgrass transplants are grown in the presence of early-growing competitive neighbors, their growth and establishment is compromised (Ross & Harper 1972; Wilson 1994; Gerry & Wilson 1995; Peltzer & Wilson 2001). Wilson and Gerry (1995) found that when restoring prairie grasses with seeding, there was a 20-fold increase in the density of native seedlings where introduced grasses had been initially treated with herbicide. Overall, these results suggest that a reduction in *C. rubescens* at or prior to planting can benefit transplants during restoration.

Although herbicide treatment allows the initial establishment of native grasses, a review of grassland restoration practices concluded that it does not provide effective long-term control of competing vegetation (Wilson 2002). Although our results indicate that the short-term survival and development of transplants can be enhanced by reducing *C. rubescens*, the longevity of this benefit remains unknown. Furthermore, even though this reduction could be achieved through a preplanting herbicide treatment, this process would also require consideration of herbicide effects on nontarget native species.

Conclusions

Using bunchgrass transplants appears to be a suitable method to assist bunchgrass restoration within recently thinned ingrown montane forests. Both *Pseudoroegneria spicata* and *Achnatherum richardsonii* were able to establish, despite drought conditions, although the use of larger transplants increased survival. Agencies practicing restoration should also consider timing planting to coincide with periods when transplant survival and growth can be maximized depending on the species. Accelerating bunchgrass establishment through transplants may allow for prompt seed production and recruitment of new bunchgrass seedlings.

Survival of transplants was positively affected by neighbor reduction of *Calamagrostis rubescens*, with limited additional benefits in tiller growth. These results suggest that an initial reduction in competitive neighboring vegetation may be useful for increasing initial transplant establishment, although this benefit may be offset by the high cost of treating such areas. A reduction in undesirable competitive vegetation may be achieved through the application of preplanting broadcast herbicide treatments, or perhaps brief periods of intensive prescribed grazing using livestock, followed by planting and prolonged rest.

Although these results indicate that bunchgrass transplants can be used to aid grassland restoration, the large-scale use of transplants may be cost prohibitive. For more widespread use of transplants to occur, establishment success must ultimately outweigh the financial cost of developing and installing transplants. Additionally, longer-term studies are required to assess the effect of neighbor cover and biomass beyond 1 year of establishment, including the recovery of neighbor vegetation following treatment and its effect on bunchgrass survival and performance.

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